Week5 Class Problems Projection

15 problems with worked solutions

Approx. 90 minutes: allow discussion and board work.

Each problem slide has intentional blank space; advance for the solution.

Problem 1: Row and column pictures

Solve $\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$. Describe both geometric pictures.

Work here before advancing.

Solution 1: Row and column pictures

Eliminate: add the equations to obtain $3x = 6$, so $x = 2$, $y = 3$. Column check: $2\begin{bmatrix}2\\1\end{bmatrix} + 3\begin{bmatrix}-1\\1\end{bmatrix} = \begin{bmatrix}1\\5\end{bmatrix}$. Two lines meet at $(2, 3)$.

Problem 2: Three outcomes

Classify $\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ c \end{bmatrix}$ for $c = 4$ and $c = 5$.

Work here before advancing.

Solution 2: Three outcomes

$R_2 \leftarrow R_2 - 2R_1$ gives $[0 \quad 0 \mid c-4]$. For $c = 4$ infinitely many $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2-t \\ t \end{bmatrix}$; for $c = 5$ no solution.

Problem 3: Find a multiplier

Eliminate below the first pivot in $\left[\begin{array}{cc|c} 2 & 3 & 7 \\ 8 & 13 & 29 \end{array}\right]$. Identify both pivots.

Work here before advancing.

Solution 3: Find a multiplier

$\ell_{21} = 8/2 = 4$. Then $R_2 \leftarrow R_2 - 4R_1$ gives $\left[\begin{array}{cc|c} 2 & 3 & 7 \\ 0 & 1 & 1 \end{array}\right]$. Pivots: $2, 1$.

Problem 4: Back substitution

Solve $\begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \\ 8 \end{bmatrix}$.

Work here before advancing.

Solution 4: Back substitution

$z = 2$, $3y + 4 = 7 \Rightarrow y = 1$, $2x + 1 - 2 = 3 \Rightarrow x = 2$. Thus $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$.

Problem 5: Full elimination

Solve $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 1 & 10 \\ 1 & 2 & 3 & 14 \end{array}\right]$ by elimination.

Work here before advancing.

Solution 5: Full elimination

$R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - R_1$ yield $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -2 \\ 0 & 1 & 2 & 8 \end{array}\right]$. Then $R_3 \leftarrow R_3 - R_2$ gives $3z = 10$. Thus $z = 10/3$, $y = 4/3$, $x = 4/3$.

Problem 6: Identify the elementary matrix

Find E implementing $R_3 \leftarrow R_3 - 4R_1$ on a 3×3 matrix.

Work here before advancing.

Solution 6: Identify the elementary matrix

$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$. Its inverse has $+4$ in position $(3, 1)$.

Problem 7: Multiply EA

Let $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$, $A = \begin{bmatrix} 3 & 1 \\ 6 & 5 \end{bmatrix}$. Compute EA and explain the operation.

Work here before advancing.

Solution 7: Multiply EA

$EA = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$. It subtracts twice row 1 from row 2.

Problem 8: The transformed right side

For $A = \begin{bmatrix} 3 & 1 \\ 6 & 5 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 4 \\ 17 \end{bmatrix}$, use $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$ to solve.

Work here before advancing.

Solution 8: The transformed right side

$E\mathbf{b} = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$, $EA = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$. So $y = 3$, $3x + 3 = 4 \Rightarrow x = 1/3$.

Problem 9: A zero pivot

Solve $\begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$ by a row exchange. Find P.

Work here before advancing.

Solution 9: A zero pivot

$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Then $PA = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$, $P\mathbf{b} = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$. Thus $y = 2$, $x = 5/3$.

Problem 10: A three-row permutation

Find P that exchanges rows 1 and 3 of $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 4 & 0 & 2 \end{bmatrix}$. Compute PA.

Work here before advancing.

Solution 10: A three-row permutation

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \text{ and } PA = \begin{bmatrix} 4 & 0 & 2 \\ 0 & 1 & 3 \\ 1 & 2 & 0 \end{bmatrix}. \text{ Also } P^{-1} = P.$$

Problem 11: Singular but consistent

Solve $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$.

Work here before advancing.

Solution 11: Singular but consistent

Row 2 is twice row 1. From $y + z = 2$, set $z = t$, $y = 2 - t$. Then $x + 2(2 - t) + t = 3$ gives $x = t - 1$. Infinitely many solutions $\begin{bmatrix} t - 1 \\ 2 - t \\ t \end{bmatrix}$.

Problem 12: Singular and inconsistent

Change the previous right side to $\mathbf{b} = \begin{bmatrix} 3 \\ 7 \\ 2 \end{bmatrix}$. Is the system solvable?

Work here before advancing.

Solution 12: Singular and inconsistent

$R_2 \leftarrow R_2 - 2R_1$ produces $[0\ 0\ 0 \mid 1]$. Contradiction: no solution.

Problem 13: Column interpretation of consistency

For $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, which vectors $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ permit $A\mathbf{x} = \mathbf{b}$?

Work here before advancing.

Solution 13: Column interpretation of consistency

The second row is twice the first, so $b_2 = 2b_1$. Equivalently both columns are multiples of $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$; $\mathbf{b}$ must lie on that line.

Problem 14: Order of matrix operations

Let $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$ and $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Compare EP and PE.

Work here before advancing.

Solution 14: Order of matrix operations

$EP = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$, while $PE = \begin{bmatrix} -2 & 1 \\ 1 & 0 \end{bmatrix}$. They differ: row operations do not generally commute.

Problem 15: Challenge: design a system

Construct a 3×3 system that needs a row swap, has three pivots, and has solution $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

Work here before advancing.

Solution 15: Challenge: design a system

One choice: $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $\mathbf{b} = A \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$. Swap rows 1 and 2; then all pivots are 1.